Palak Verma

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Summary

Computer Science Engineering student with hands-on experience in Machine Learning, NLP, LLM applications, FastAPI, and full-stack development. Built AI-powered healthcare, recommendation, and resume screening systems using Python, Scikit-Learn, Pandas, NumPy, and vector search technologies. Strong foundation in Data Structures, Algorithms, DBMS, OS, and backend engineering.

Education

B.Tech in Computer Science Engineering 2024–2028
Bennett University, Greater Noida
CGPA: 8.33

Class XII (2024): 91% **Class X (2022):** 97%
St. Thomas College, Lucknow

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript
Frontend/Backend: React.js, HTML5, CSS3, Next.js, FastAPI, REST API, Node.js
AI/ML: Machine Learning, NLP, Vector Embeddings, FAISS, RAG,
HuggingFace Transformers, Scikit-Learn, Pandas, NumPy
Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP
Databases: MySQL, PostgreSQL, MongoDB
Tools: Git, GitHub, Streamlit, Vercel, Railway

Work Experience

AI/ML Development Intern — Katharos Techie | GitHub Jun 2026 – Jul 2026
- Engineered AI-powered commerce analytics solutions using Python, FastAPI, LLMs, recommendation systems, and semantic search technologies.
- Designed and deployed backend workflows for KPI summarization, anomaly detection, and conversational analytics to support intelligent business decision-making.

Projects

Nyay AI – Legal Aid Chatbot | GitHub | Live Demo FastAPI, React.js, FAISS, Groq, PostgreSQL
- Built a multilingual RAG-based legal aid chatbot using FAISS vector search, Cohere reranking, and LLaMA 3.1-8B via Groq, with JWT-authenticated FastAPI backend and PostgreSQL.
- Added voice I/O via faster-whisper and gTTS; deployed full stack on Railway and Vercel.

Resume Analyzer | GitHub Python, Streamlit
- Built an AI-powered resume analyzer using NLP, TF-IDF, cosine similarity, and hybrid semantic search.
- Designed an ATS scoring engine with 90%+ ranking consistency for automated candidate-job matching.

Crop Recommendation System ML, Next.js
- Developed a crop recommendation system using Machine Learning to predict suitable crops based on soil and climate conditions.
- Trained a Random Forest model on 2,200+ agricultural records, achieving 87% prediction accuracy.

Leadership & Extracurricular Activities

Design Team Head — RPA Club
- Led team delivering assets for 5+ technical events
- Mentored junior designers and maintained branding consistency

Student Council Sub-Head — IT Committee
- Coordinated 10+ college events and supported digital initiatives
- Assisted in website updates and execution

Certifications

- Data Structures & Algorithms — Infosys Springboard
- Machine Learning using Python — Great Learning
- Core Java Programming — Great Learning / Wingspan
- Database Systems — Saylor Academy
- Python Programming — Great Learning
- Statistics — Saylor Academy